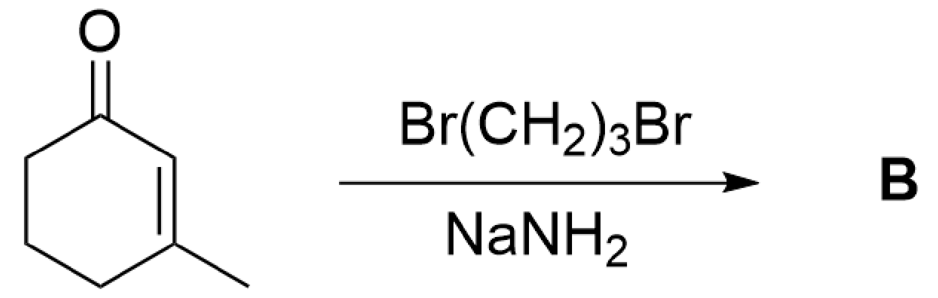
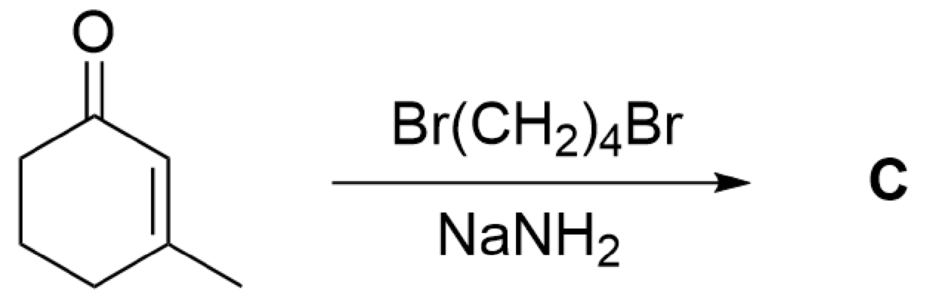
题目



O=C1C=C(C)CCC1>BrCCCBr.[Na]N>[B],B是产物

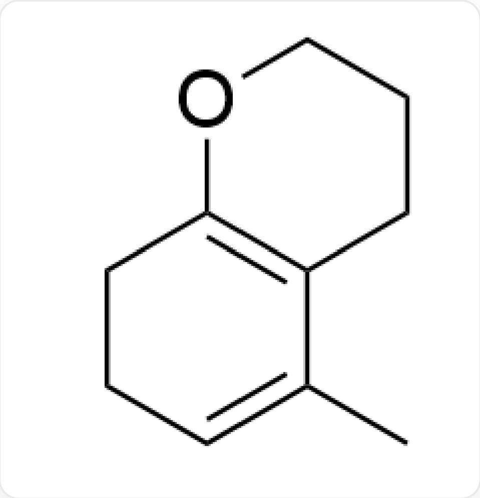


O=C1C=C(C)CCC1>BrCCCCBr.[Na]N>[C],C是产物

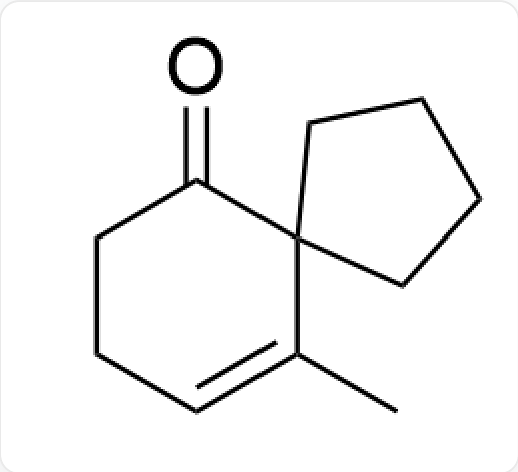
已知在氨基钠的液氨溶液中，反应得到的产物B和C略有不同，在不考虑对映异构的条件下，试分别给出产物B和C的结构式

A. 其他选项均不正确

B.

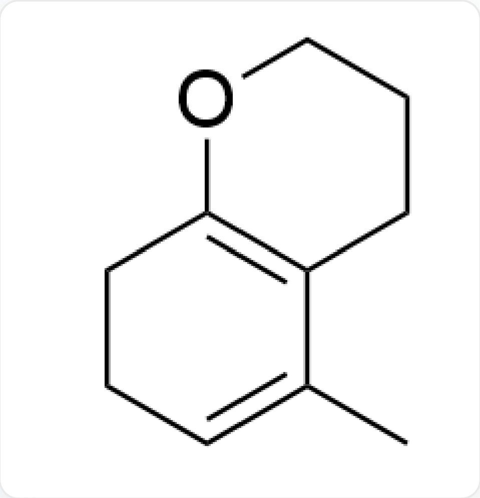


CC1=CCCC2=C1CCCO2

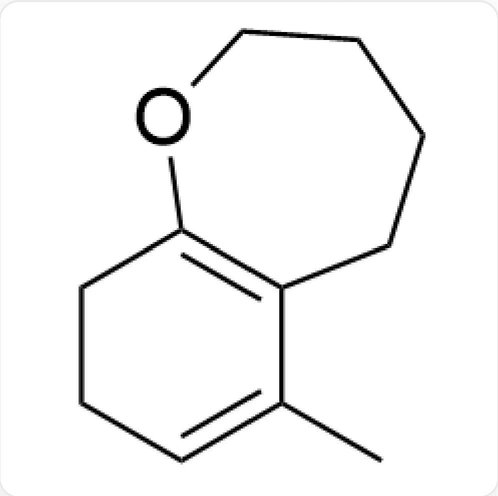


O=C1C2(CCCC2)C(C)=CCC1

C.

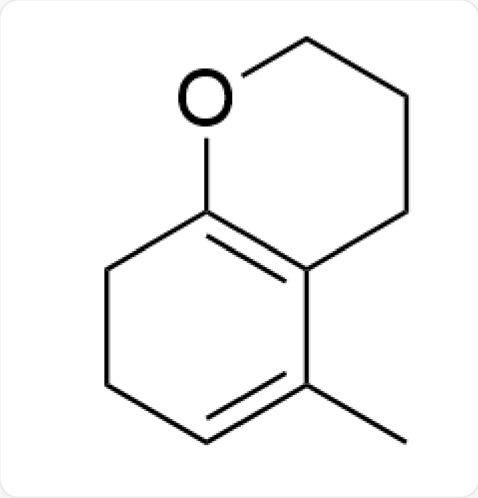


CC1=CCCC2=C1CCCO2

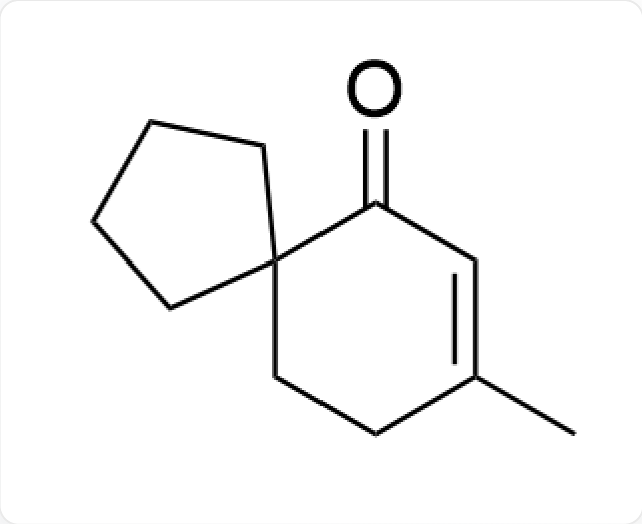


CC1=CCCC2=C1CCCCO2

D.

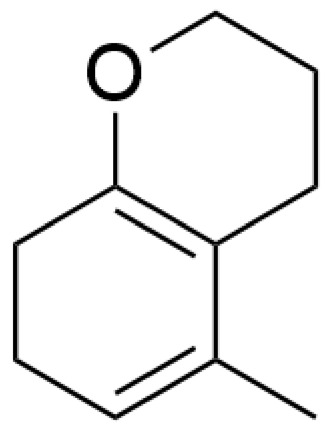


CC1=CCCC2=C1CCCO2

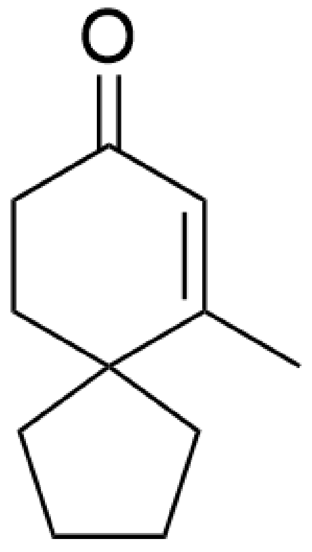


O=C1C=C(C)CCC12CCCC2

E.

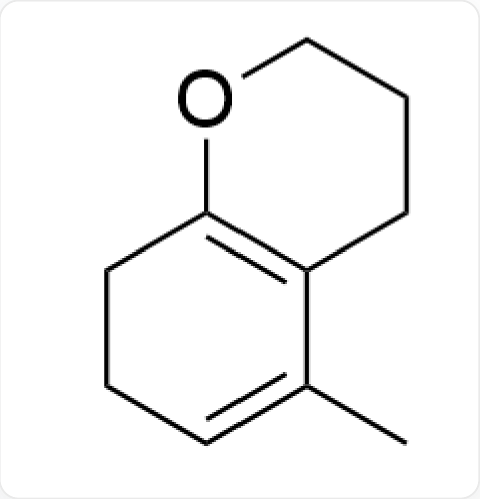


CC1=CCCC2=C1CCCO2

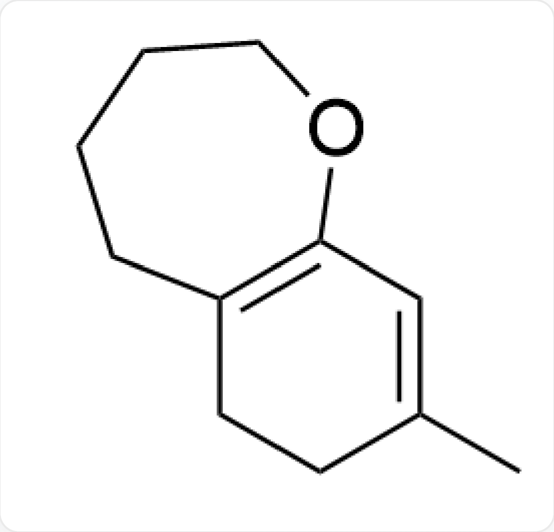


O=C1C=C(C)C2(CCCC2)CC1

F.

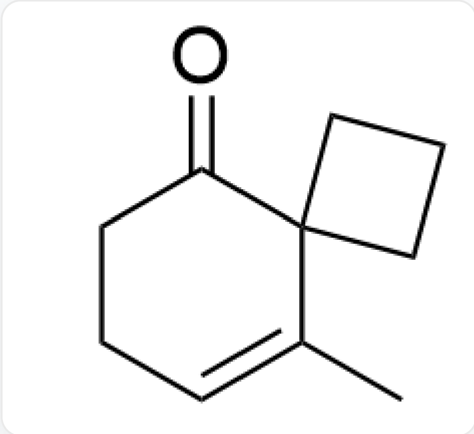


CC1=CCCC2=C1CCCO2

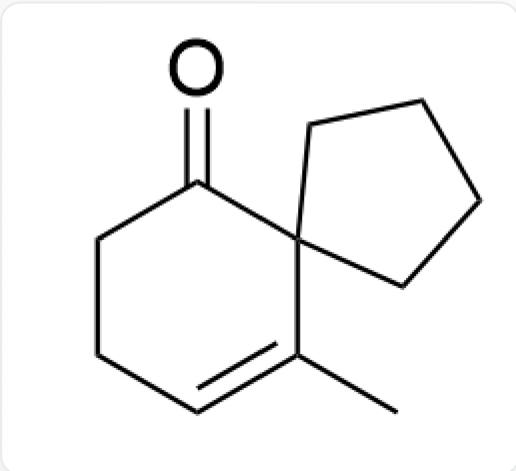


CC1=CC(OCCCC2)=C2CC1

G.

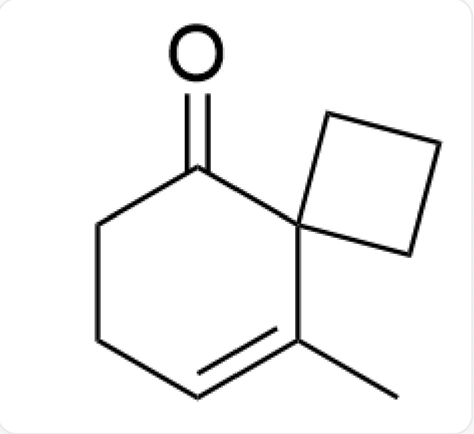


O=C1C2(CCC2)C(C)=CCC1

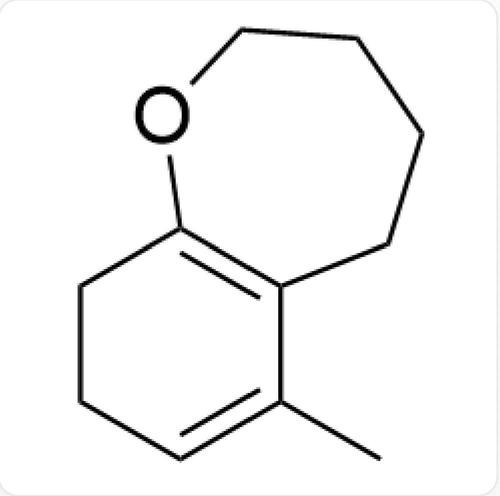


O=C1C2(CCCC2)C(C)=CCC1

H.

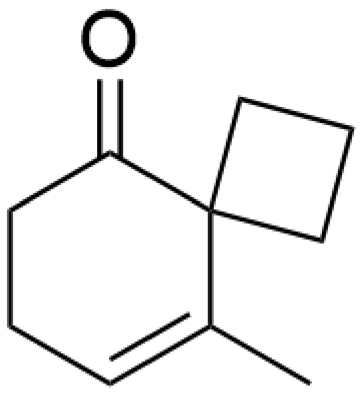


O=C1C2(CCC2)C(C)=CCC1

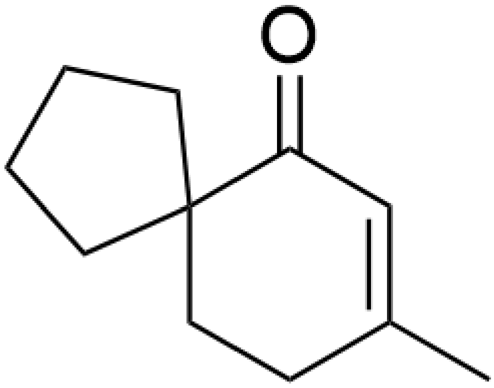


CC1=CCCC2=C1CCCCO2

I.

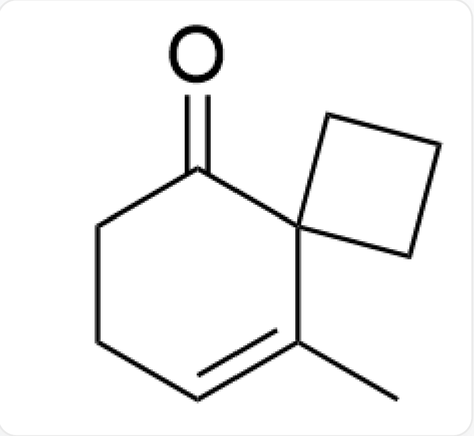


O=C1C2(CCC2)C(C)=CCC1

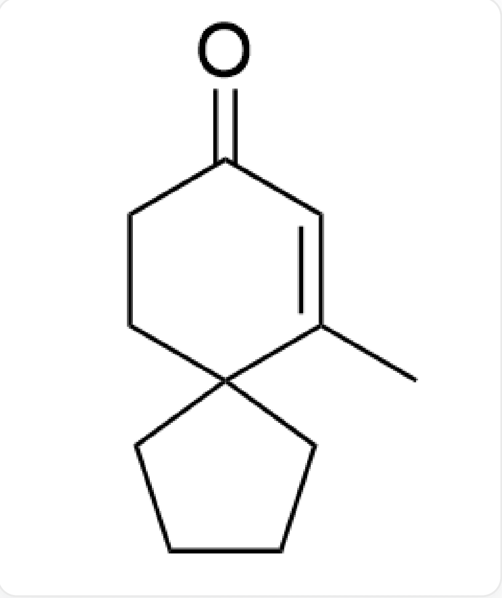


O=C1C=C(C)CCC12CCCC2

J.

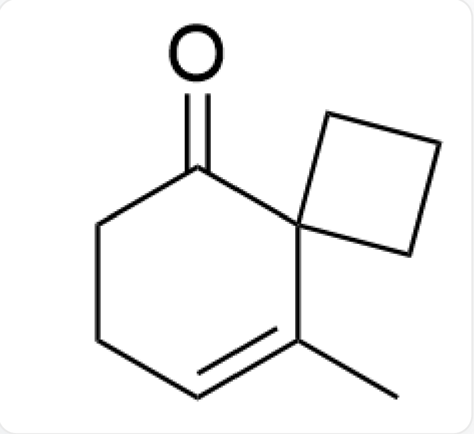


O=C1C2(CCC2)C(C)=CCC1

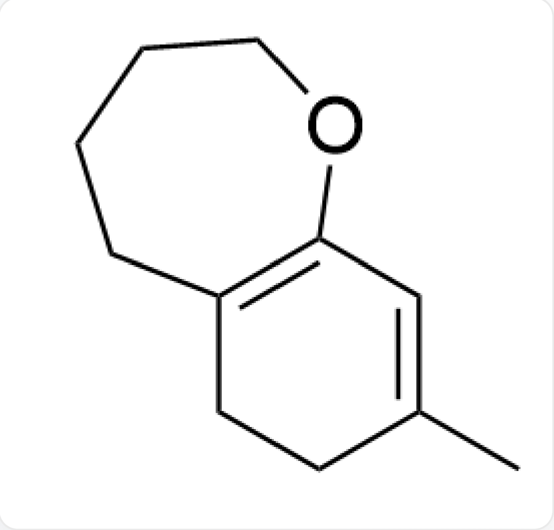


O=C1C=C(C)C2(CCCC2)CC1

K.

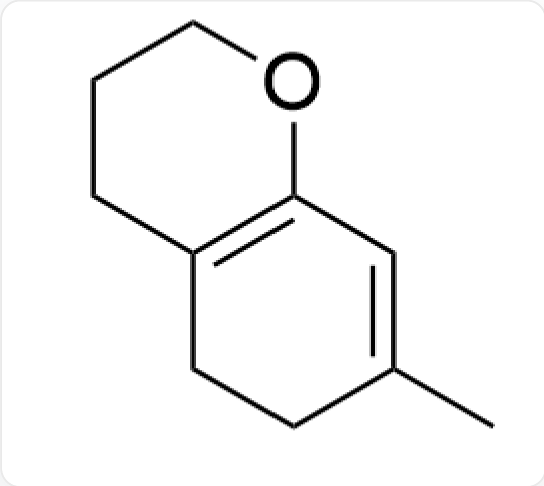


O=C1C2(CCC2)C(C)=CCC1

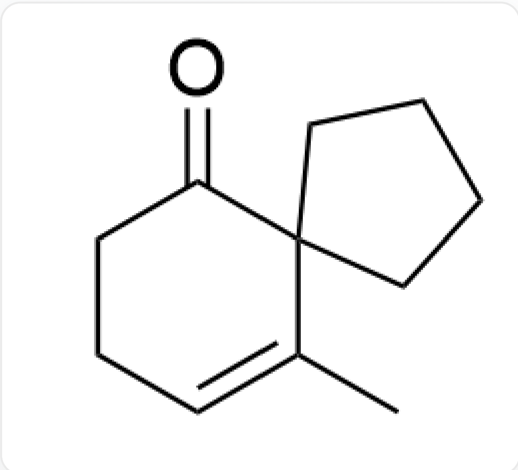


CC1=CC(OCCCC2)=C2CC1

L.

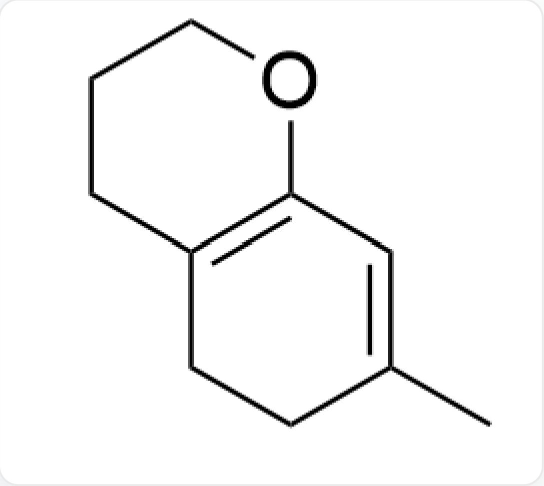


CC1=CC(OCCC2)=C2CC1

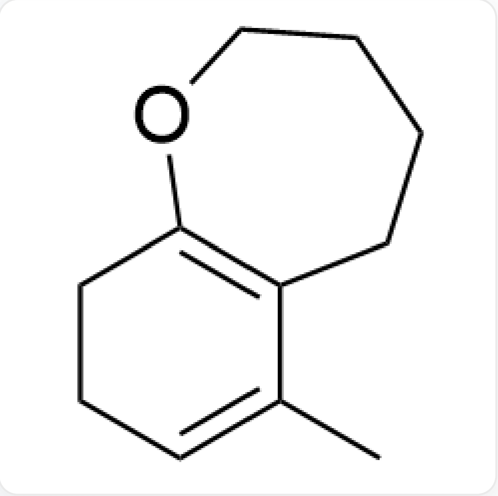


O=C1C2(CCCC2)C(C)=CCC1

M.

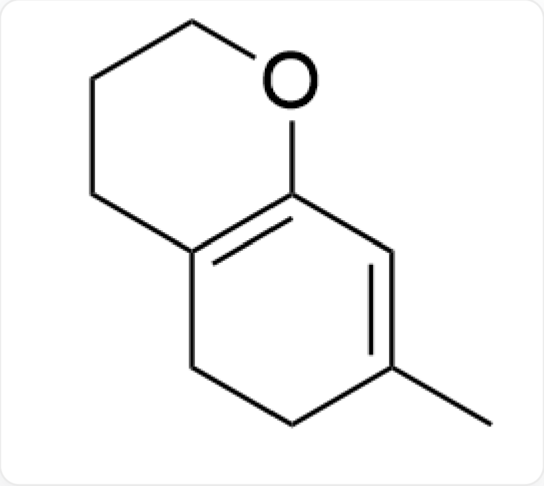


CC1=CC(OCCC2)=C2CC1

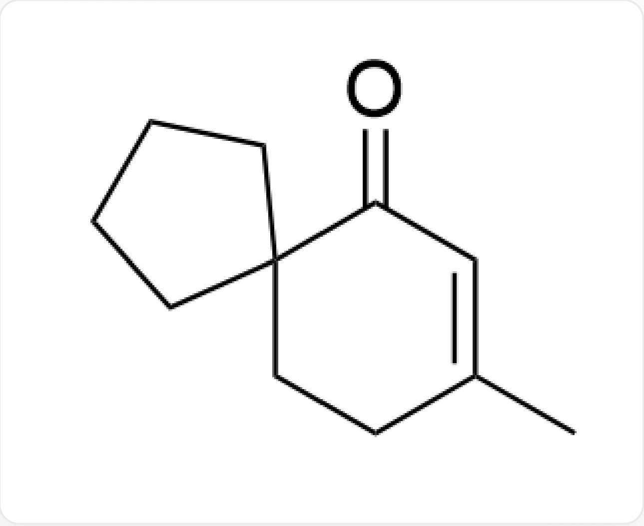


CC1=CCCC2=C1CCCCO2

N.

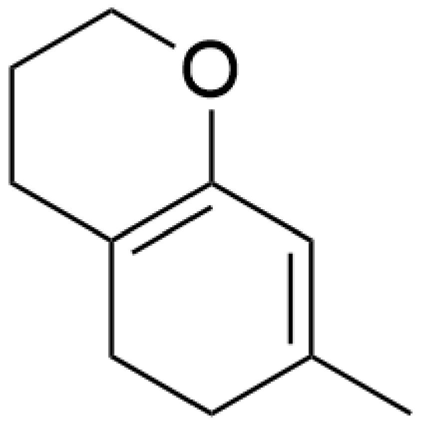


CC1=CC(OCCC2)=C2CC1

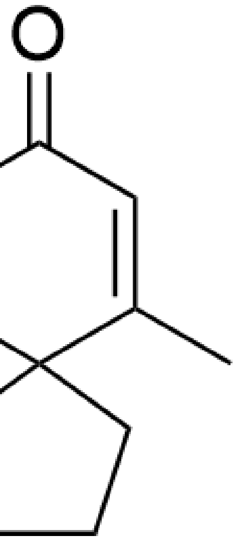


O=C1C=C(C)CCC12CCCC2

O.

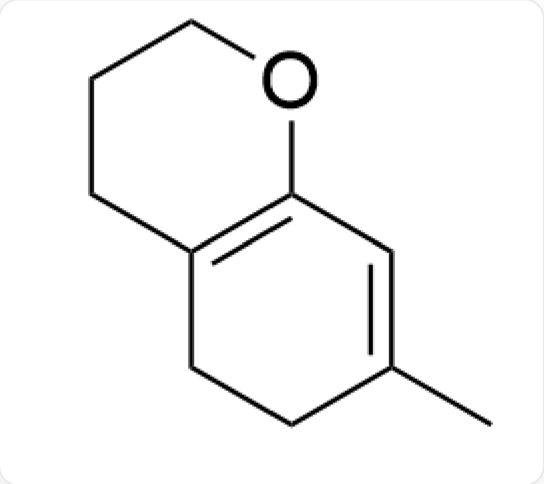


CC1=CC(OCCC2)=C2CC1

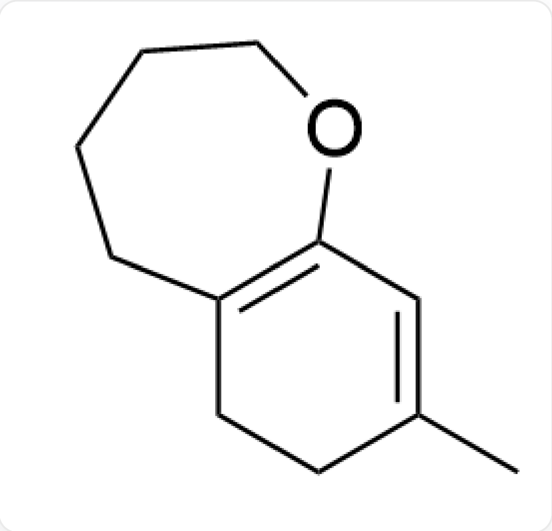


O=C1C=C(C)C2(CCCC2)CC1

P.



CC1=CC(OCCC2)=C2CC1



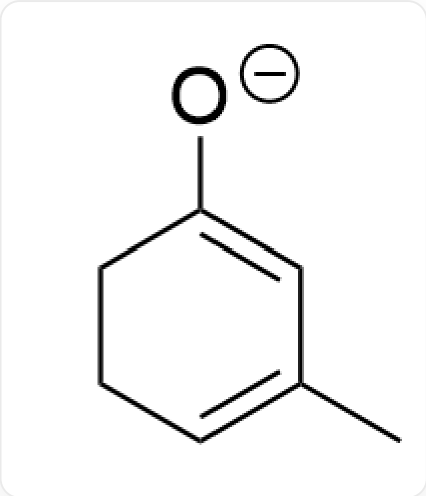
CC1=CC(OCCCC2)=C2CC1

答案

正确答案: B

详细解析

首先，在液氨体系中，倾向于优先形成热力学稳定的线性共轭阴离子中间体 **1**



[O-]C1=CC(C)=CCC1

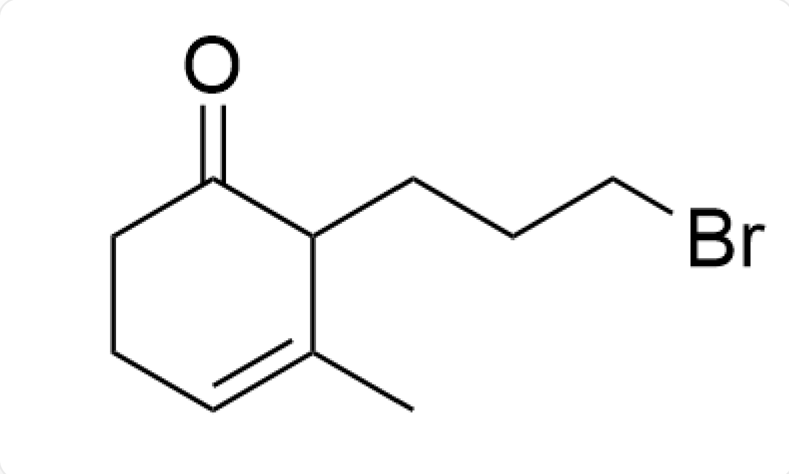
CHECKPOINT 1 PTS

阴离子中间体 **1** :[O-]C1=CC(C)=CCC1

对于生成产物 **B** 和 **C** 的反应，由于碳负离子在羰基 α 位更加稳定，因此首先都与相应的卤代烷反应分别得到中间体 **2** 和中间体 **3**

CHECKPOINT 1 PTS

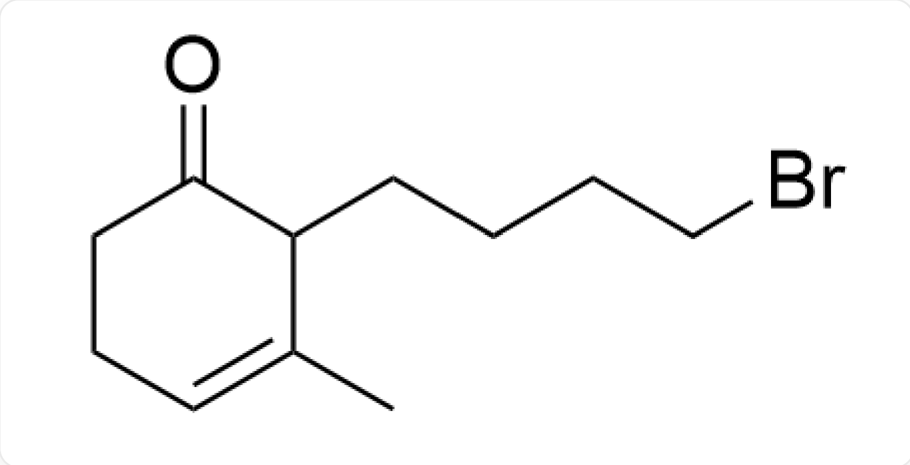
由于碳负离子在羰基α位更加稳定，因此首先都与相应的卤代烷反应分别得到中间体**2**和中间体**3**



中间体**2**：O=C1C(CCCBr)C(C)=CCC1

CHECKPOINT 1 PTS

中间体**2**： O=C1C(CCCBr)C(C)=CCC1

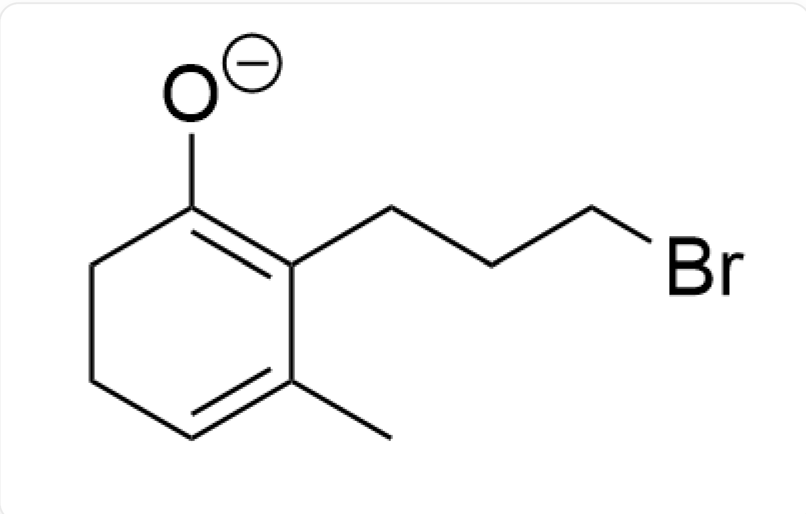


中间体**3**：O=C1C(CCCCBr)C(C)=CCC1

CHECKPOINT 1 PTS

中间体 **3**： O=C1C(CCCCBr)C(C)=CCC1

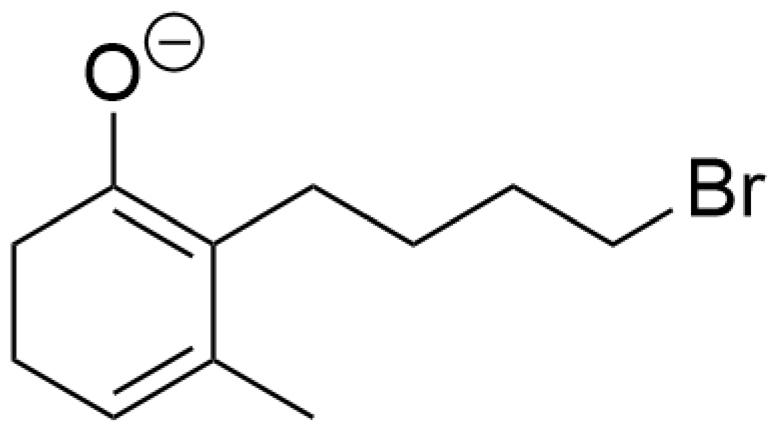
接着再被碱攫取一个质子分别得到中间体 **4** 和中间体 **5**



中间体**4**：[O-]C1=C(CCCBr)C(C)=CCC1

CHECKPOINT 1 PTS

中间体**4**：[O-]C1=C(CCCBr)C(C)=CCC1



中间体**5**：[O-]C1=C(CCCCBr)C(C)=CCC1

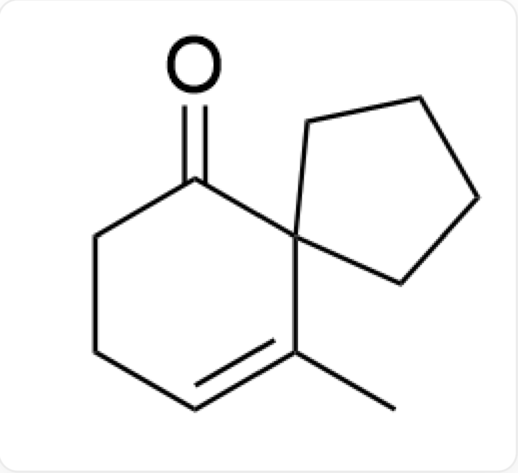
CHECKPOINT 1 PTS

中间体**5**：[O-]C1=C(CCCCBr)C(C)=CCC1

对于中间体**5**，由于碳端亲核能力强于氧端，因此优先得到含五元环的产物**C**

CHECKPOINT 1 PTS

对于中间体**5**，由于碳端亲核能力强于氧端，因此优先得到含五元环的产物**C**



产物**C**：O=C1C2(CCCC2)C(C)=CCC1

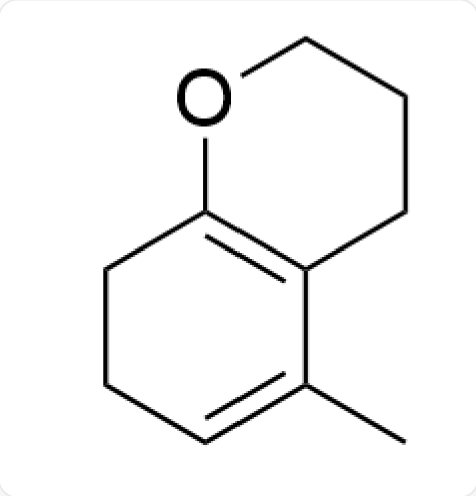
CHECKPOINT 1 PTS

产物 **C**： O=C1C2(CCCC2)C(C)=CCC1

而对于中间体 **4**，由于碳端亲核形成的四元环不稳定，因此倾向于用氧端亲核，形成含两个六元环的产物 **B**

CHECKPOINT 1 PTS

而对于中间体 **4**，由于碳端亲核形成的四元环不稳定，因此倾向于用氧端亲核，形成含两个六元环的产物 **B**



产物**B**：CC1=CCCC2=C1CCCO2

CHECKPOINT 1 PTS

产物**B**：CC1=CCCC2=C1CCCO2